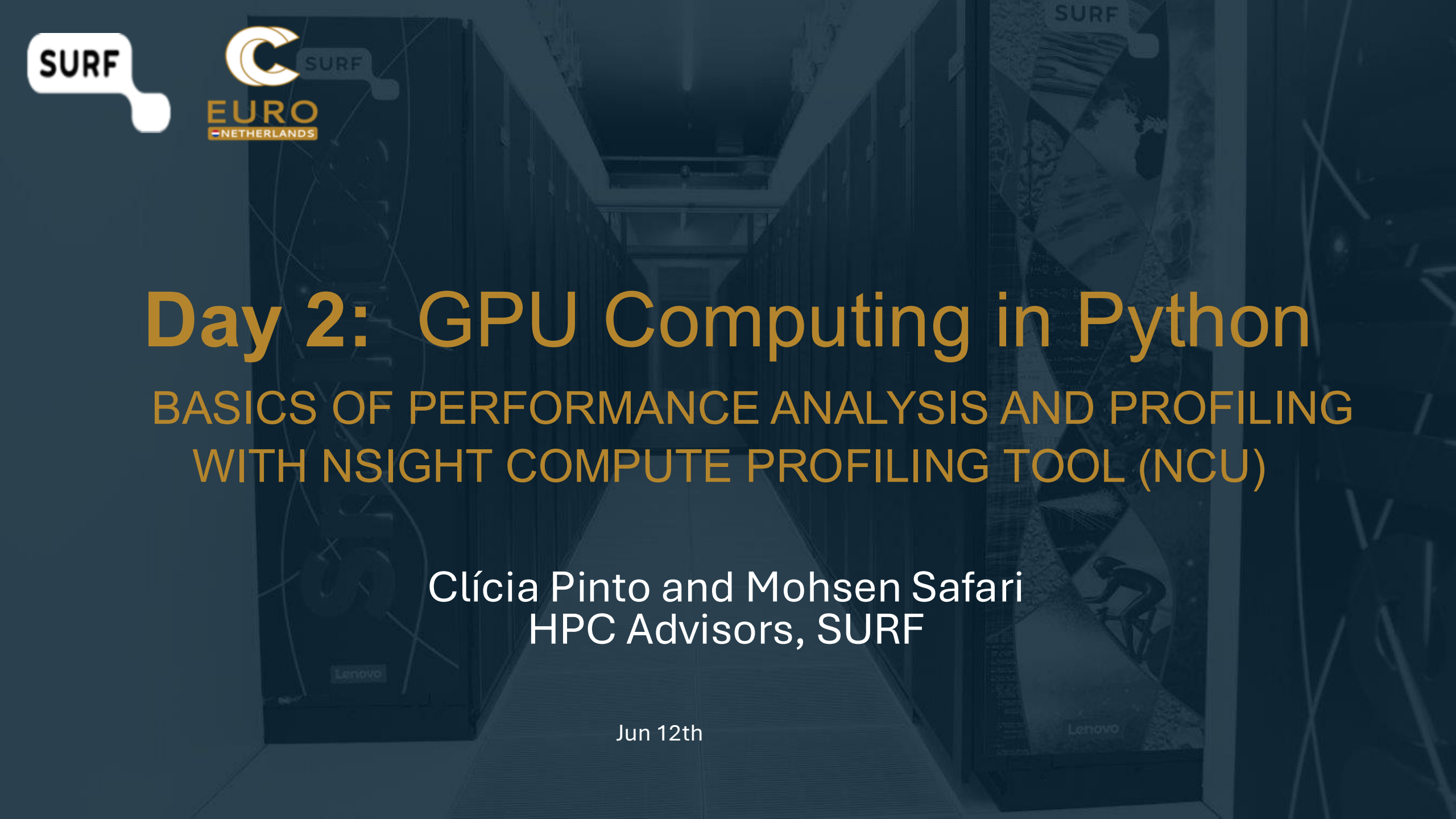


Day 2: GPU Computing in Python

BASICS OF PERFORMANCE ANALYSIS AND PROFILING WITH NSIGHT COMPUTE PROFILING TOOL (NCU)

Clícia Pinto and Mohsen Safari
HPC Advisors, SURF

Jun 12th

Amsterdam Science Park

Who we are

SURF Expertise Pool:

- High-Performance Computing
- High-Performance Machine Learning
- High-performance visualization
- Quantum Computing
- Data preservation
- Data management
- Cloud for Research
- System administration
- Trust and Identity
- Software Management

Outline

- Profiling Tool - What it is and does
- Nsight Compute Profiling Tool (ncu)
- Profiling Applications on Snellius

Resources

- **Slides and source code of the examples can be found at:**
  - https://github.com/sara-nl/Parallel-and-GPU-programming-in-Python/tree/main/Day2
- **Programming environment (jupyter):**
  - https://ondemand.snellius.surf.nl/
- **To reset your password and Accept the User Agreement:**
  - https://portal.cua.surf.nl
- **To have more information on Snellius partitions and hardware:**
  - https://servicedesk.surf.nl/wiki/spaces/WIKI/pages/30660209/Snellius+partitions+and+accounting

Profiling Tool

What it is and does

- Characterizes hardware **performance metrics**
- Detects **performance issues**
- Supports **analysis** workflows
- Enables **baseline comparisons** across GPU architectures for cross-platform development
- Shows how performance relates to **underlying hardware concepts**
- Helps users identify **potential bottlenecks** and **underutilization**
- Correlates **efficiency** and **performance** metrics down to individual **lines of code**

Profiling Tool

What it is and does

Performance:

- **How fast** a task is completed by the computer
  - Execution time
  - Throughput
  - Bandwidth
  - FLOPS

Efficiency:

- **How well** the computational resources are used to achive a given performance
  - Resource utilization
  - % of peak performance

Profiling Tool

What it is and does

- Performance Analysis

- Characterizes your workflow in
  - **Memory-bound**
    - Memory throughput is high
  - **Compute-bound**
    - Compute throughput is high

Profiling Tool

What it is and does

GPU Performance:

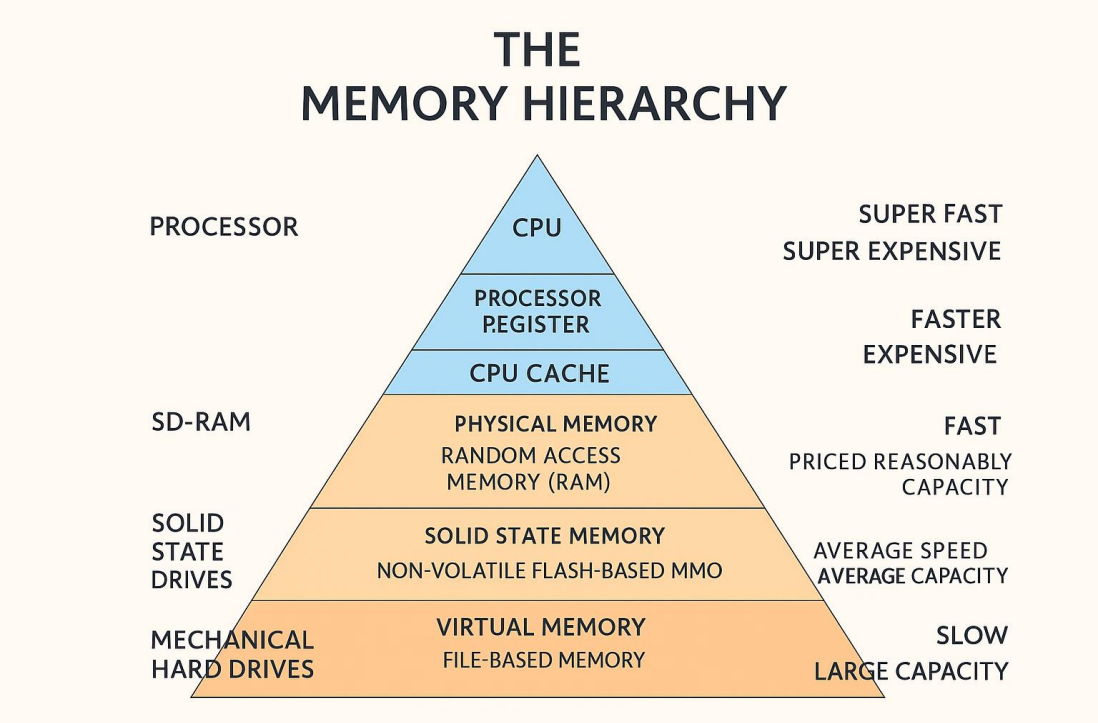
- **Unoptimized GPU programs are expensive** in terms of time, power and money
- Optimized GPU programs need to take advantage of several **execution units**
- High **trade-off between** development **time** and optimization **effort**

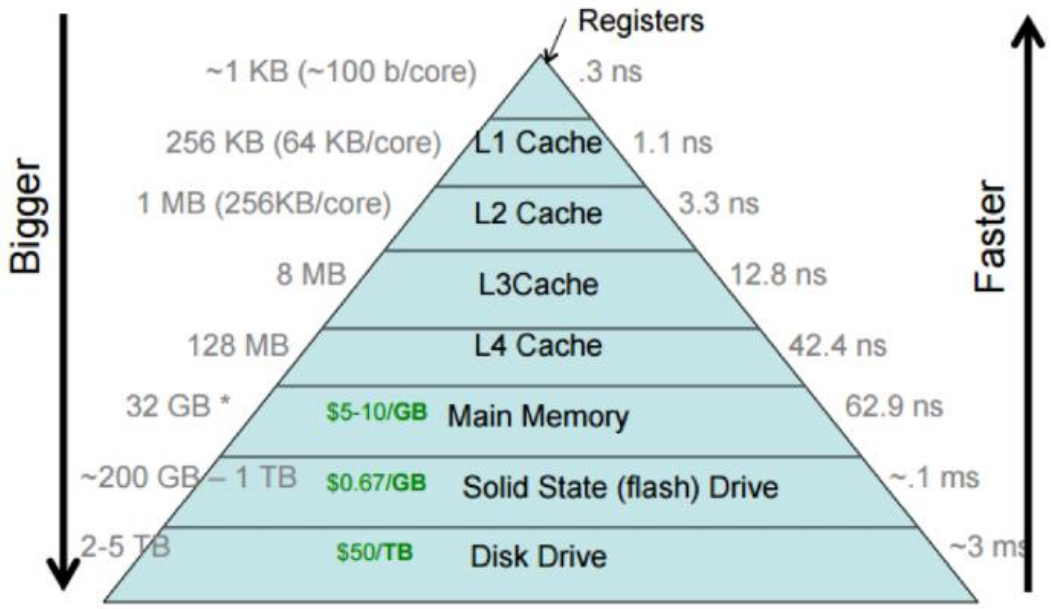
Think about:

- What is the purpose of using GPUs, and how is this code expected to evolve over time?
- What do I need in terms of performance? Fast execution times? Better throughput? Better occupancy of the GPU node?

Profiling Tool

Memory Hierarchy





Profiling Tool

Which tasks are best candidates for GPU?

✅ **Best** suited for GPU acceleration:

- High data parallelism
- Tasks that perform the same operations on many independent data elements.
- High computational intensity and low memory intensity
- Large amounts of computation per byte transferred.
- Examples include:
- Dense linear algebra
- Matrix–matrix multiplication
- Simulation kernels
- FFTs
- Monte Carlo methods
- Convolutions
- Physics simulations

❌ **Less** suitable for GPUs:

- Memory-heavy applications
- Workloads dominated by data movement rather than computation
- Frequent, small I/O operations
- Especially between host and device or network I/O.
- Irregular memory access patterns
- Unpredictable or scattered data access reduces GPU efficiency.
- Significant branching or divergence
- Heavy use of if–else, switch, or other divergent control flow.

Nsight Compute Profiling Tool (ncu)

What it is

- Interactive **kernel profiler** for CUDA applications
- Identifies **performance** and **efficiency** bottlenecks
- Analyses the device **memory usage**
- Translates metric values into **actionable** information
- It provides:
  - Detailed **performance metrics**
  - API **debugging**

- Can be used through **user interface (UI)** and **command line (CLI)** tool
  - (This course will cover the command line tool part)
- Differs from Nsight System (Nsys) profiler that provides **system-wide** profiling
- Ncu offers **kernel-based** profiling

SURF

Nsight Compute Profiling Tool (ncu)

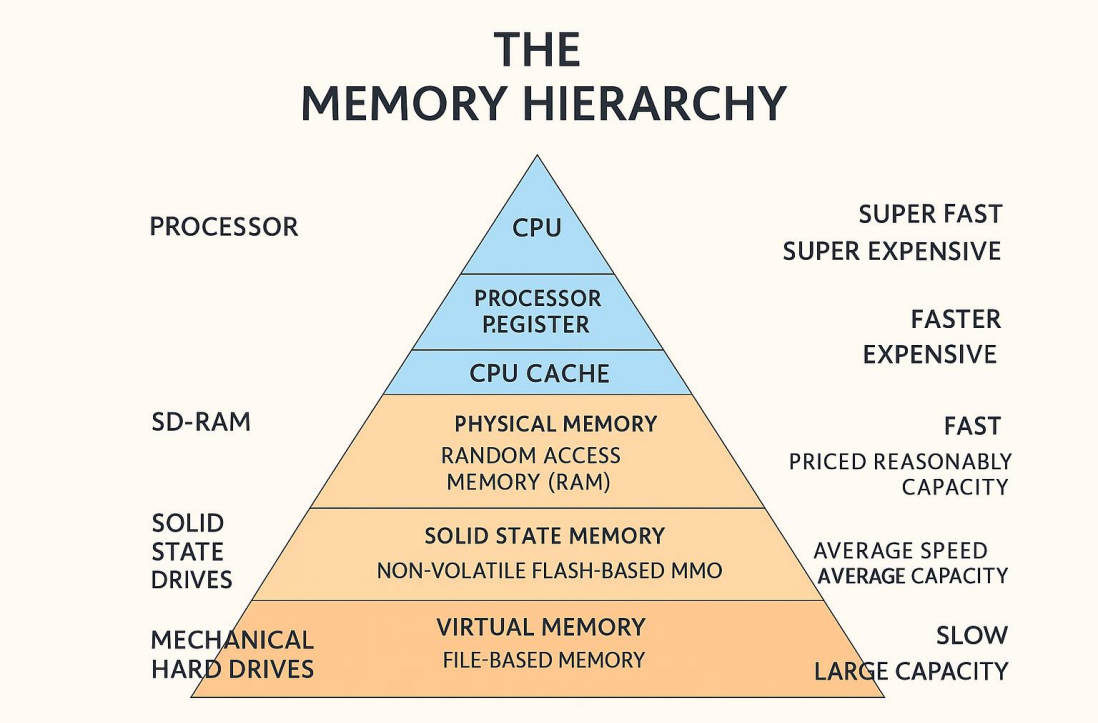
Where to find

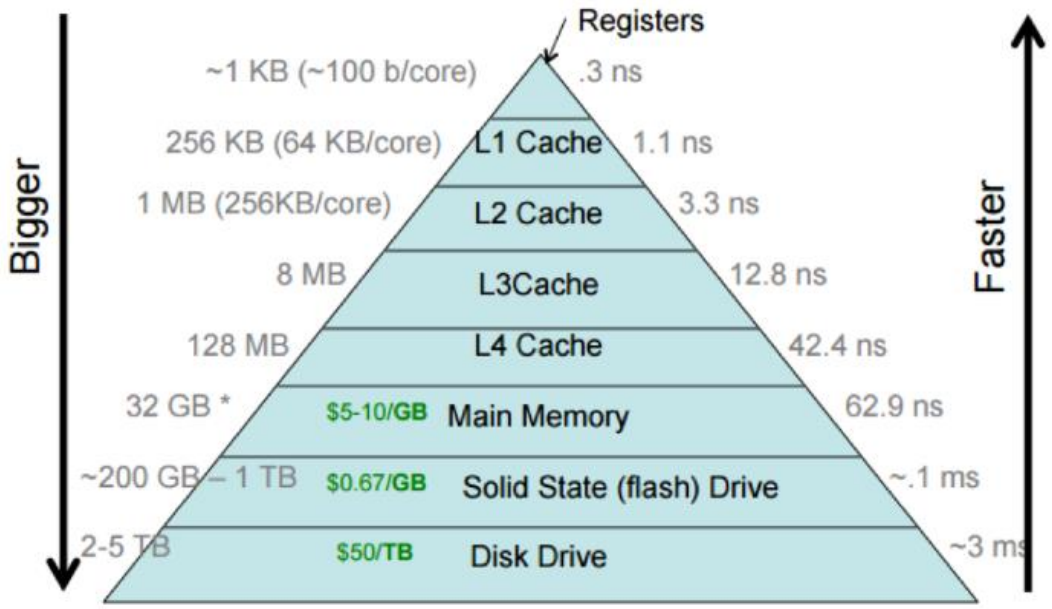
- Is available under the regular CUDA Toolkit installation
- On Snellius:

```
module load 2025
module load CUDA/12.8.0
```

Profiling Tool

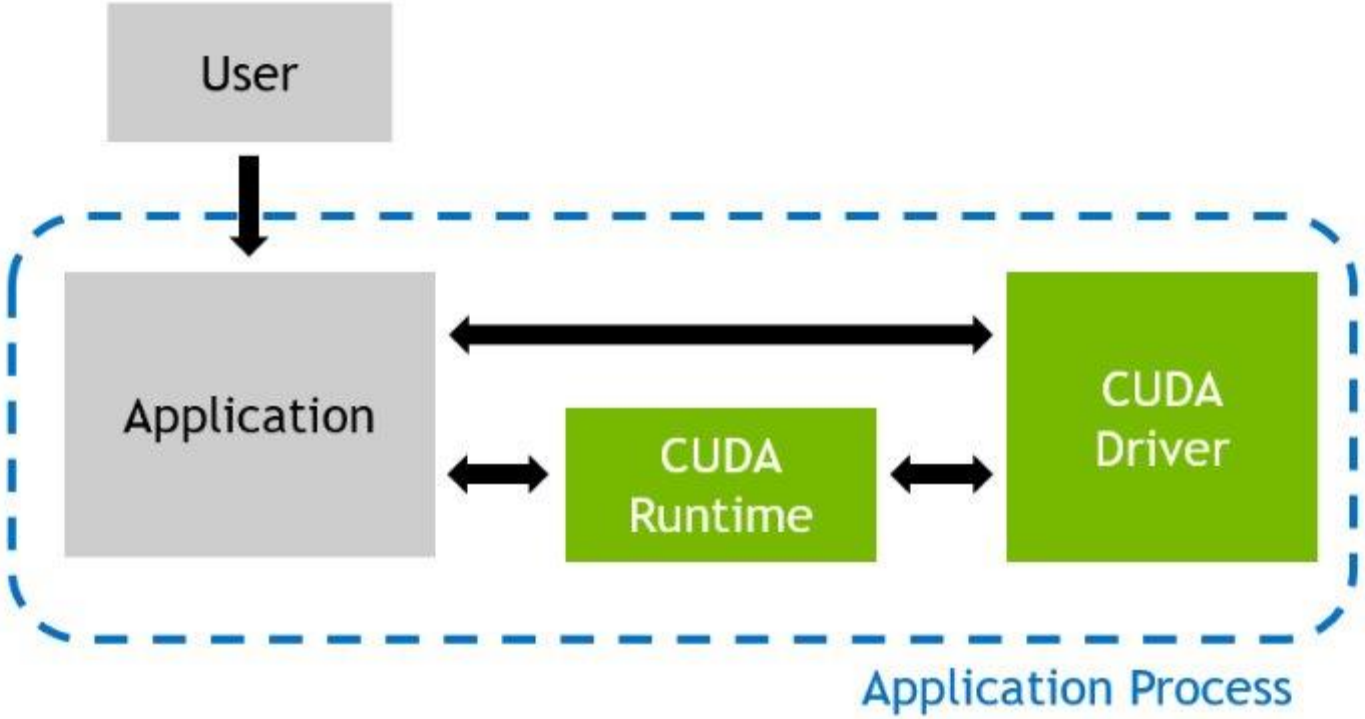
Memory Hierarchy





Nsight Compute Profiling Tool (ncu)

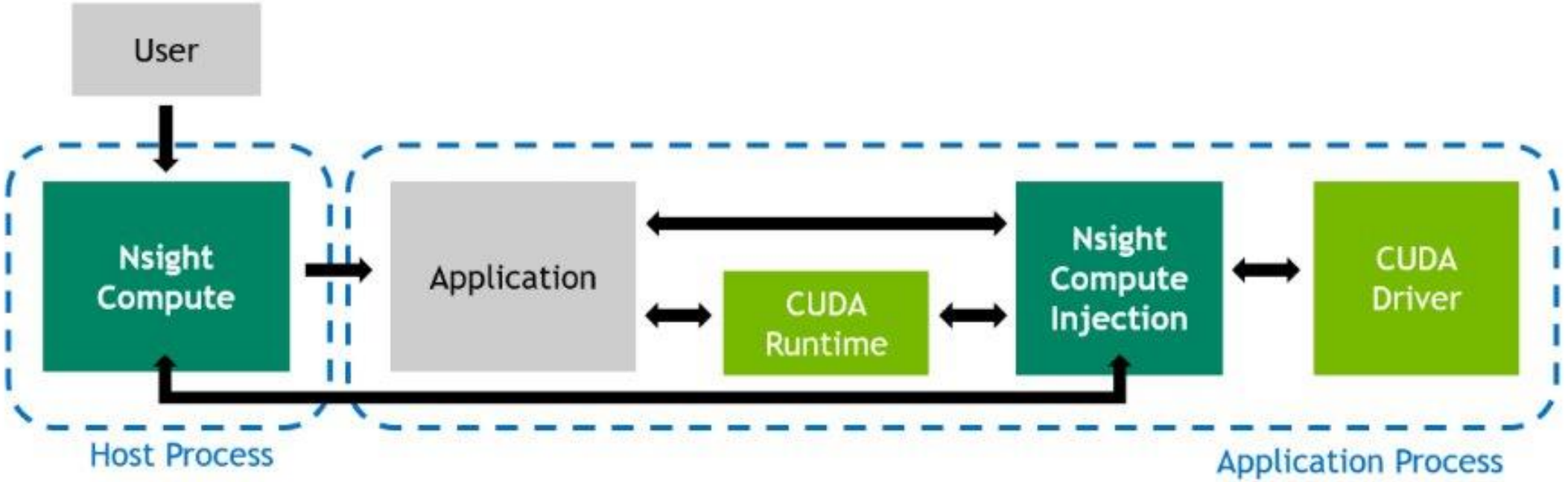
Profiling Applications



Regular Application Execution

Nsight Compute Profiling Tool (ncu)

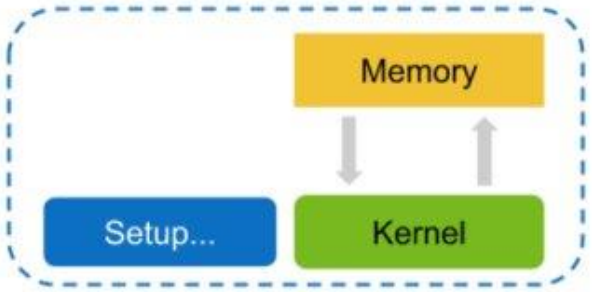
Profiling Applications



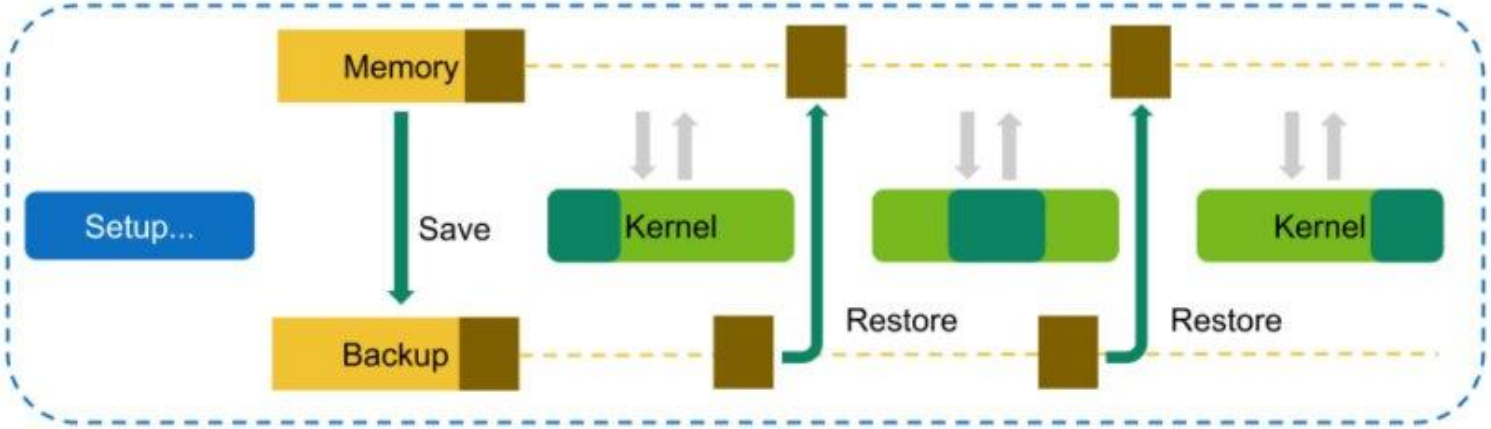
Profiled Application Execution

Nsight Compute Profiling Tool (ncu)

Metric Collection: Kernel Replay



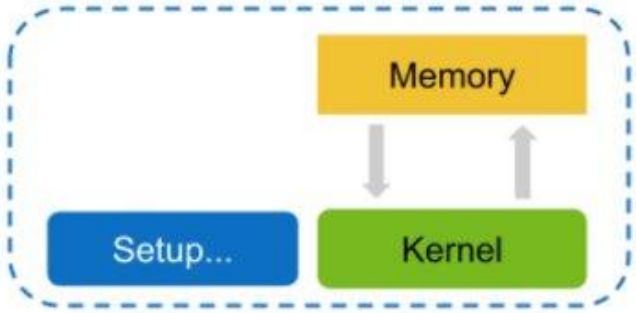
Regular Application Execution



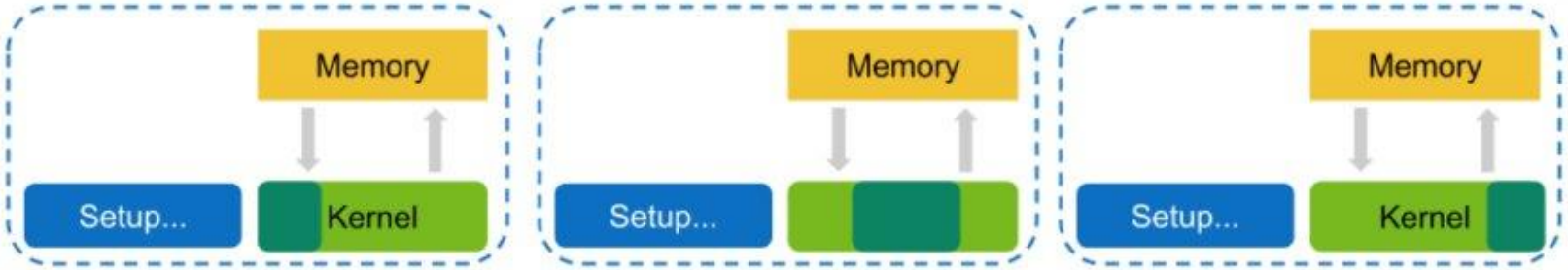
Execution with Kernel Replay. All memory is saved, and memory written by the kernel is restored in-between replay passes.

Nsight Compute Profiling Tool (ncu)

Metric Collection: Application Replay



Regular Application Execution



Execution with Application Replay. No memory is saved or restored, but the cost of running the application itself is duplicated.

Nsight Compute Profiling Tool (ncu)

Metric Collection: Sets and Sections

- Commonly used subset of metrics are grouped into pre-defined sets or sections
- Each set and section added adds also an overhead to the whole collection

Identifier
ComputeWorkloadAnalysis (Compute Workload Analysis)
InstructionStats (Instruction Statistics)
LaunchStats (Launch Statistics)
MemoryWorkloadAnalysis (Memory Workload Analysis)
NUMA Affinity (NumaAffinity)
Nvlink (Nvlink)
Nvlink_Tables (Nvlink_Tables)
Nvlink_Topology (Nvlink_Topology)
PM Sampling (PmSampling)
PM Sampling: Warp States (PmSampling_WarpStates)
SchedulerStats (Scheduler Statistics)
SourceCounters (Source Counters)
SpeedOfLight (GPU Speed Of Light Throughput)
WarpStateStats (Warp State Statistics)

Nsight Compute Profiling Tool (ncu)

Metric Collection: Default

- Default collection:
  - High-level utilization
  - Static Launch
  - Occupancy

Identifier
ComputeWorkloadAnalysis (Compute Workload Analysis)
LaunchStats (Launch Statistics)
MemoryWorkloadAnalysis (Memory Workload Analysis)
SpeedOfLight (GPU Speed Of Light Throughput)
WarpStateStats (Warp State Statistics)
Occupancy (Occupancy)

Nsight Compute Profiling Tool (ncu)

Example 1: vector-add-v2.py (N = 100,000)

- **Basic collection:**
  - ncu python 3-vector-add-v0.py

```
cliciad@gcn30 basic_examples]$ ncu –o output.txt python 1-vector-add-v0.py

==PROF== Connected to process 466370 (/gpfs/admin/_hpc/sw/arch/INTEL-
AVX512/RHEL9/EB_production/2024/software/Python/3.12.3-GCCcore-13.3.0/bin/python3.12)
==PROF== Profiling "addition" - 0: 0%....50%....100% - 10 passes
Elapsed on GPU with PyCuda (sec):  1.367883544921875
--------------------
Elapsed time using CPU sequential for-loop (sec):  4.398094415664673
--------------------
Validation: there are 0 different element(s)!
--------------------
==PROF== Disconnected from process 1858485
[466370] python3.13@127.0.0.1
  addition (19532, 1, 1)x(512, 1, 1), Context 1, Stream 7, Device 0, CC 8.0
```

Nsight Compute Profiling Tool (ncu)

Example 1: vector-add-v2.py (N = 100,000)

- **Basic collection:**
  - ncu python 3-vector-add-v0.py

```
cliciad@gcn30 basic_examples]$ ncu –o output.txt python 1-vector-add-v0.py

==PROF== Connected to process 466370 (/gpfs/admin/_hpc/sw/arch/INTEL-
AVX512/RHEL9/EB_production/2024/software/Python/3.12.3-GCCcore-13.3.0/bin/python3.12)
==PROF== Profiling "addition" - 0: 0%....50%....100% - 10 passes
Elapsed on GPU with PyCuda (sec):  1.367883544921875
--------------------
Elapsed time using CPU sequential for-loop (sec):  4.398094415664673
--------------------
Validation: there are 0 different element(s)!
--------------------
==PROF== Disconnected from process 1858485
[466370] python3.13@127.0.0.1
  addition (19532, 1, 1)x(512, 1, 1), Context 1, Stream 7, Device 0, CC 8.0
```

Nsight Compute Profiling Tool (ncu)

Example 1: vector-add-v2.py (N = 100,000)

```
Section: GPU Speed Of Light Throughput
---------------------- ----------- ------------
Metric Name            Metric Unit Metric Value
---------------------- ----------- ------------
DRAM Frequency                 Ghz         1.20
SM Frequency                   Ghz         1.08
Elapsed Cycles               cycle        4,428
Memory Throughput                %        12.76
DRAM Throughput                  %        12.76
Duration                        us         4.10
L1/TEX Cache Throughput          %         7.81
L2 Cache Throughput              %        14.60
SM Active Cycles             cycle     2,111.67
Compute (SM) Throughput          %         3.24
---------------------- ----------- ------------
```

```
Section: Launch Statistics
------------------------------ -------------- --------------
Metric Name                        Metric Unit    Metric Value
------------------------------ -------------- --------------
Block Size                                                512
Function Cache Configuration                  CachePreferNone
Grid Size                                                 196
Registers Per Thread             register/thread           16
Shared Memory Configuration Size           Kbyte        16.38
Driver Shared Memory Per Block        Kbyte/block         1.02
Dynamic Shared Memory Per Block       byte/block            0
Static Shared Memory Per Block        byte/block            0
# SMs                                         SM          108
Stack Size                                             1,024
Threads                                   thread      100,352
# TPCs                                                     54
Enabled TPC IDs                                          all
Uses Green Context                                         0
Waves Per SM                                            0.45
------------------------------ -------------- --------------
```

```
Section: Occupancy
------------------------------ ----------- ------------
Metric Name                    Metric Unit Metric Value
------------------------------ ----------- ------------
Block Limit SM                       block           32
Block Limit Registers                block            8
Block Limit Shared Mem               block           16
Block Limit Warps                    block            4
Theoretical Active Warps per SM       warp           64
Theoretical Occupancy                    %          100
Achieved Occupancy                       %        42.42
Achieved Active Warps Per SM          warp        27.15
------------------------------ ----------- ------------
```

```
Section: GPU and Memory Workload Distribution
------------------------- ----------- ------------
Metric Name               Metric Unit Metric Value
------------------------- ----------- ------------
Average DRAM Active Cycles      cycle       627.10
Total DRAM Elapsed Cycles       cycle      198,656
Average L1 Active Cycles        cycle     2,111.91
Total L1 Elapsed Cycles         cycle      499,844
Average L2 Active Cycles        cycle     1,814.34
Total L2 Elapsed Cycles         cycle      344,160
Average SM Active Cycles        cycle     2,111.91
Total SM Elapsed Cycles         cycle      499,844
Average SMSP Active Cycles      cycle     2,204.97
Total SMSP Elapsed Cycles       cycle    1,999,376
------------------------- ----------- ------------
```

```
OPT   Est. Local Speedup: 57.58%
      The difference between calculated theoretical (100.0%) and measured achieved occupancy (42.4%) can be the
      result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can
      occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices
      Guide (https://docs.nvidia.com/cuda/cuda-c-best-practices-guide/index.html#occupancy) for more details on
      optimizing occupancy.
```

```
OPT   If you execute __syncthreads() to synchronize the threads of a block, it is recommended to have at least two
      blocks per multiprocessor (compared to the currently executed 1.8 blocks) This way, blocks that aren't
      waiting for __syncthreads() can keep the hardware busy.

OPT   This kernel grid is too small to fill the available resources on this device, resulting in only 0.5 full
      waves across all SMs. Look at Launch Statistics for more details.
```

SURF

Nsight Compute Profiling Tool (ncu)

Example 1: vector-add-v2.py (N = 100,000)

```
OPT   Est. Speedup: 5.41%
      One or more SMs have a much lower number of active cycles than the average number of active cycles. Maximum
      instance value is 11.86% above the average, while the minimum instance value is 27.51% below the average.
----- ---------------------------------------------------------------------------------------------------------
OPT   Est. Speedup: 6.235%
      One or more SMSPs have a much lower number of active cycles than the average number of active cycles. Maximum
      instance value is 13.09% above the average, while the minimum instance value is 28.62% below the average.
----- ---------------------------------------------------------------------------------------------------------
OPT   Est. Speedup: 5.41%
      One or more L1 Slices have a much lower number of active cycles than the average number of active cycles.
      Maximum instance value is 11.86% above the average, while the minimum instance value is 27.51% below the
      average.
----- ---------------------------------------------------------------------------------------------------------
OPT   Est. Speedup: 12.21%
      One or more L2 Slices have a much higher number of active cycles than the average number of active cycles.
      Maximum instance value is 28.96% above the average, while the minimum instance value is 9.33% below the
      average.
```

Nsight Compute Profiling Tool (ncu)

Example 1: vector-add-v2.py (N = 100,000,000)

```
  Section: GPU Speed Of Light Throughput
  ----------------------- ----------- ------------
  Metric Name             Metric Unit Metric Value
  ----------------------- ----------- ------------
  DRAM Frequency                  Ghz         1.21
  SM Frequency                    Ghz         1.09
  Elapsed Cycles                cycle      951,181
  Memory Throughput                 %        88.28
  DRAM Throughput                   %        88.28
  Duration                         us       868.70
  L1/TEX Cache Throughput           %        16.20
  L2 Cache Throughput               %        69.27
  SM Active Cycles              cycle   944,526.70
  Compute (SM) Throughput           %        12.21
  ----------------------- ----------- ------------
```

```
Section: Occupancy
------------------------------ ----------- ------------
Metric Name                    Metric Unit Metric Value
------------------------------ ----------- ------------
Block Limit SM                       block           32
Block Limit Registers                block            8
Block Limit Shared Mem               block           16
Block Limit Warps                    block            4
Theoretical Active Warps per SM       warp           64
Theoretical Occupancy                    %          100
Achieved Occupancy                       %        85.51
Achieved Active Warps Per SM          warp        54.73
------------------------------ ----------- ------------
```

```
Section: Launch Statistics
-------------------------------- --------------- ---------------
Metric Name                          Metric Unit    Metric Value
-------------------------------- --------------- ---------------
Block Size                                                   512
Function Cache Configuration                      CachePreferNone
Grid Size                                                195,313
Registers Per Thread               register/thread              16
Shared Memory Configuration Size                Kbyte           16.38
Driver Shared Memory Per Block        Kbyte/block            1.02
Dynamic Shared Memory Per Block       byte/block               0
Static Shared Memory Per Block        byte/block               0
# SMs                                         SM             108
Stack Size                                                 1,024
Threads                                   thread     100,000,256
# TPCs                                                       54
Enabled TPC IDs                                             all
Uses Green Context                                            0
Waves Per SM                                             452.11
-------------------------------- --------------- ---------------
```

```
OPT   Est. Local Speedup: 14.49%
      The difference between calculated theoretical (100.0%) and measured achieved occupancy (85.5%) can be the
      result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can
      occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices
      Guide (https://docs.nvidia.com/cuda/cuda-c-best-practices-guide/index.html#occupancy) for more details on
      optimizing occupancy.
```

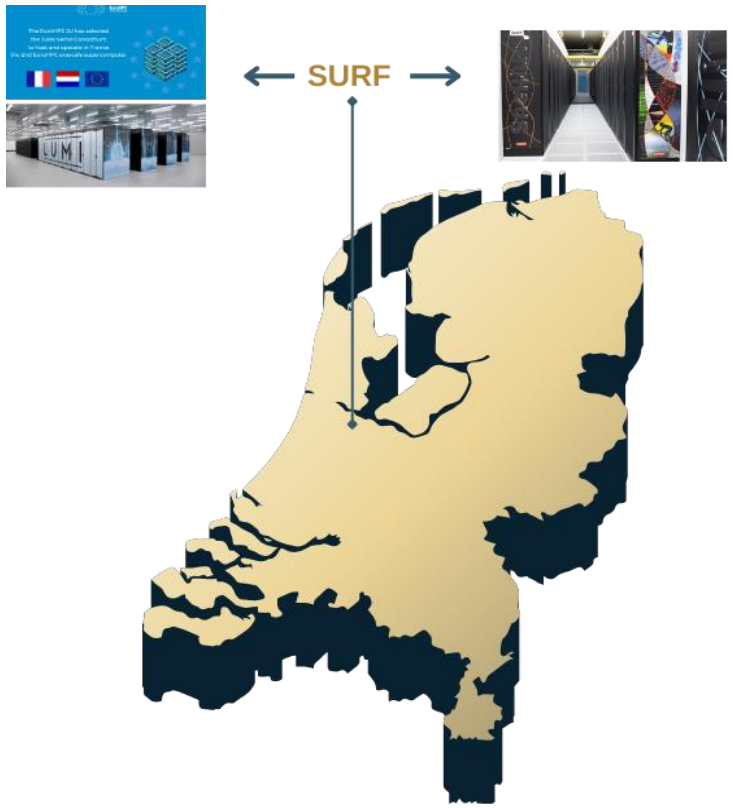
```
INF   This workload is utilizing greater than 80.0% of the available compute or memory performance of the device.
      To further improve performance, work will likely need to be shifted from the most utilized to another unit.
      Start by analyzing DRAM in the Memory Workload Analysis section.
```

Thank You

Funded by the European Union. This work has received funding from the European High Performance Computing Joint Undertaking (JU) and Germany, Bulgaria, Austria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, Greece, Hungary, Ireland, Italy, Lithuania, Latvia, Poland, Portugal, Romania, Slovenia, Spain, Sweden, France, Netherlands, Belgium, Luxembourg, Slovakia, Norway, Türkiye, Republic of North Macedonia, Iceland, Montenegro, Serbia under grant agreement No 101101903.

Access to Supercomputers and GPUs

Current



Future

- Alice Recoque Exascale System (France - 2026)

  The Alice Recoque supercomputer will be hosted and operated by the Jules Verne consortium, led by France through GENCI and CEA, with the participation of the Netherlands through SURF.

- AI Factory Groningen - 2027

  consortium partners SURF, TNO and Samenwerking Noord, has taken the initiative for this facility and is working on its implementation, aiming to have it fully operational by 2027.

Supercomputers
EuroHPC Systems

PROCURED

5 PETASCALE

- Vega in Slovenia
- Karolina in Czechia
- Discoverer in Bulgaria
- Meluxina in Luxembourg
- Deucalion in Portugal

3 PRE-EXASCALE

- Lumi in Finland
- Leonardo in Italy
- Marenostrum 5 in Spain

ONGOING

2 EXASCALE

- Jupiter in Germany
- Alice Recoque in France

2 MID-RANGE

- Arrhenius in Sweden
- Daedalus in Greece

COMING NEXT

UPGRADES

- Discoverer+
- Lisa/Leonardo

AN INDUSTRIAL SYSTEM

- Co-owned and for use by the industrial sector
- For AI and other applications

A POST-EXASCALE SYSTEM

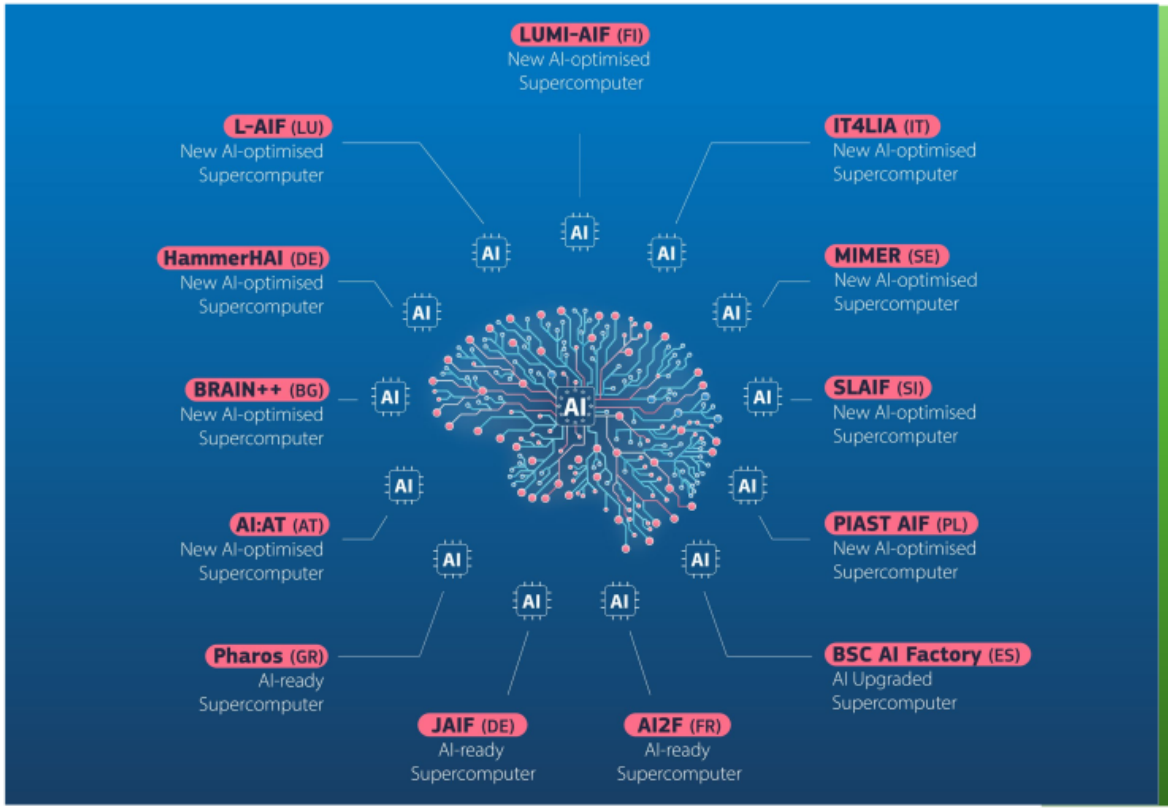
PROCUREMENT OF FEDERATION SERVICES

- A platform for the federation of EuroHPC HPC and quantum infrastructure
- A one-stop shop access point for users

Supercomputers
AI factories (EuroHPC Systems)

THE AI FACTORIES

EuroHPC JU selected 13 EU sites that will host the first AI Factories – to drive Europe's leadership in AI



AI-ready EuroHPC supercomputers in:

- Germany **JAIF** – JUPITER
- France **AI2F** – Alice Recoque
- Greece **Pharos** – Daedalus

AI-upgrades to EuroHPC supercomputers in:

- Spain **BSC AIF** – MareNostrum 5

New AI-optimized EuroHPC supercomputers in:

- Finland **LUMI-AIF**
- Germany **HammerHAI**
- Italy **IT4ALIA**
- Luxembourg **L-AIF**
- Sweden **MIMER**
- Bulgaria **BRAIN++**
- Slovenia **SLAIF**
- Austria **AI:AT**
- Poland **PIAST AIF**

*AI Factories pull together EU and national resources, in a collaborative effort of **21 European countries***